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DigiCAS-HPS: A tailored digital competence assessment scale for health professions students in the post-COVID era in Vietnam

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ABSTRACT

The COVID-19 pandemic has accelerated the adoption of digital health, highlighting the critical role of digital health competencies in delivering reliable and effective healthcare services. These competencies remain essential post-pandemic to integrate eHealth technologies, telemedicine, and IoT-enabled healthcare solutions into routine clinical practice. This study aimed to develop and validate a tailored digital competence assessment scale (DigiCAS-HPS) for health professions students and to identify differences in digital competence across various demographic and academic groups. A stratified sampling method selected 717 health professions students from four majors in Dong Thap province, Vietnam. Based on previous literature on digital frameworks, pilot testing, and expert review, we generated, refined, and utilized the questionnaire to collect data from the first sample of 366 students to identify the underlying latent factor structure by an exploratory factor analysis. We then applied a confirmatory factor analysis to affirm the structural validity and model-data fit with the second sample of 351 students. The Mann-Whitney and Kruskal Wallis test determined significant differences in digital competence levels among student groups classified by gender, age, field of study, and academic year. The DigiCAS-HPS scale resulted in 16 items and two factors, encompassing Factor 1 (Digital Interaction & Responsibility) and Factor 2 (Digital Content Development and Software Mastery), and demonstrated valid results in model fit, construct validity, and reliability. Personal factors such as older and soon-to-graduate students revealed significant associations with higher proficiency in digital content development and software mastery ($p < 0.05$). The DigiCAS-HPS scale might be a helpful tool for educators, healthcare institutions, and policymakers to assess and integrate digital competencies into healthcare education. To address the demands of eHealth, telemedicine, and IoT-enabled healthcare systems, the scale supports the development of a digitally proficient healthcare workforce prepared to navigate the dynamic and technology-driven healthcare landscape.
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Introduction

The COVID-19 pandemic has affected every industry and service. Healthcare activities, including routine appointments, elective procedures, non-urgent hospital visits, and general physician surgeries, were canceled. The pressure on healthcare organizations to quickly adapt forced them to use and repurpose available technologies to maintain safe communication and contact between families and healthcare professionals and enable healthcare services to be delivered remotely.[1,2] Those technical innovations require healthcare professionals' information communication technology (ICT) skills.[3,4] Healthcare educational institutions started the integration of remote care demonstrations, videoconferencing, and online modules into instructional methods to prepare health professions students working in digital environments.[5,6] A qualified digital competence scale might be a critical tool for determining the ICT knowledge and skill gaps among health professions students. Such a tool can guide the development of curriculum in healthcare education to equipe future healthcare professionals with digital competencies required to adapt to evolving technological landscaple and improve patient care.[7,8].

However, despite the growing emphasis on digital health competencies, there is a notable lack of validated tools specifically designed to assess these competencies among health professions

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students in low- and middle-income countries (LMICs), including Vietnam. [9] The development of existing instruments often takes place in high-income settings, which may not accurately reflect the contextual, educational, and technological realities of students in LMICs. [10] This misalignment can lead to poor evaluations and ineffective course planning. While the COVID-19 pandemic pushed Vietnam's healthcare and education sectors to adopt digital transformation in response to the COVID-19 pandemic rapidly, the tools to measure students' readiness and digital competence haven't kept up for this context. Therefore, there is a critical need for a reliable, culturally sensitive, and context-appropriate assessment scale to evaluate digital competencies among health professions students in Vietnam. This study aimed to develop and quantitatively validate a tailored digital competence assessment scale for health professions students (DigiCAS-HPS) in the context of the post-COVID-19 era. Additionally, it investigated how personal factors, such as gender, age, field of study, and academic year, influenced digital competence levels among students.

We proposed the following hypotheses to guide the investigation:

H1: The DigiCAS-HPS will demonstrate acceptable psychometric properties, including construct validity, internal consistency reliability, and model fit indices, supporting its use for assessing digital competence among health professions students in Vietnam.

H2: There will be significant differences in students' digital competence scores based on demographic and academic characteristics, particularly age and academic year, with older and more senior students expected to show higher competence levels.

H3: Gender and field of study will not be significantly associated with differences in digital competence scores.

Materials and methods

Design of DigiCAS-HPS instrument

The development of DigiCAS-HPS involved three phases to ensure the development of the instrument with scientific guarantees: 1) conceptualization and item pool generation, 2) content validity assessment and pilot testing, and 3) testing of construct validity and reliability (internal consistency).[11].

Phase I

The measurement (dimension/factor/component) conceptualization was determined by reviewing popular frameworks of digital competence in the world [12–14], systematic reviews [9,10,15], and qualitative studies on healthcare professionals' perceptions of digital health competence.[16,17] The inductive approach was applied to detect concepts.[18] A group of researchers and two experts in ICT generated an item pool relevant to those concepts. Each item was crafted to fit the intended construct. The measurement type is the Likert type scaling that was used to indicate the level of likelihood of every item as follows: 1 point (Very unlikely: I have never done it or I can't do it even with someone's help); 2 points (Unlikely: I can't do it myself without help); 3 points (Not confirmed: Sometimes I can do it myself without help); 4 points (Likely: I can do it on my own and solve problems that arise); 5 points (Very likely: I can master myself, and solve problems, and teach others at the same time).

Phase II: Content validity assessment

Content validity assessment determined whether the initial item pool sufficiently represented the phenomenon or proposed domains.[19] A panel of six experts was selected via purposive sampling based on their experience (≥5 years) in healthcare, health informatics, education, or instrument development. All had prior involvement in tool validation or digital health education and agreed to independently review the instrument. They independently reviewed the initial pool of items for relevance, clarity, and representativeness. The item level content validity index (I-CVI) was calculated for each item, with a minimum acceptable threshold of 0.83 [20]. The scale-level content validity index (S-CVI) was computed, with a benchmark of ≥ 0.85 indicating acceptable overall content validity [20]. Expert feedback was qualitatively reviewed for clarity, terminology, and domain representation. Items with I-CVI values below this threshold were revised or removed. Suggestions from multiple experts regarding item wording, duplication, or alignment with digital health constructs were integrated by consensus among the research team to refine the item pool for subsequent pilot testing and factor analysis. A pilot study was conducted to assess the feasibility and face validity of the instruments by asking ten students and gathering feedback on the technical functionality, such as readability, clarity, and completion time of the scale items.[21].

Phase III: Construct validation and reliability assessment

Exploratory factor analysis (EFA) with principle axis factoring and Promax rotation were employed to identify latent factors of the first version of DigiCAS-HPS.[22,23] Kaiser-Meyer-Olkin (KMO) test assessed the sampling adequacy, and Bartlett's sphericity (BTOS) test ensured that variables in the population correlation matrix were correlated.[23] Factor retention was based on eigenvalues > 1.0, visual inspection of the scree plot, and theoretical interpretability [23]. Items with factor loading values higher than 0.65 were allocated to the respective factors in the rotated factor matrix and no substantial cross-loadings (> 0.30 on multiple factors).[24] Cronbach's alpha coefficient was used to assess the internal consistency reliability of the instrument's items using a threshold of ≥ 0.70 for newly developed instruments.[25] These analyses were conducted using IBM® SPSS software version 20.0 (SPSS Inc, Chicago, IL, USA).

CFA was performed using IBM® SPSS AMOS version 20 to verify the factor structure and test model fit. Model fit was assessed using multiple indices to provide a comprehensive evaluation: Chi-square/df, RMSEA, and SRMR for absolute and residual fit; CFI and TLI for incremental fit; and GFI for overall model adequacy. These indices were selected based on their widespread use and interpretability in structural equation modeling literature [26,27]. The following benchmarks of fit indices are as follows: Chi-square/df < 3.0, Root Mean Square Error of Approximation (RMSEA) ≤ 0.06, Standardized Root Mean Square Residual (SRMR) ≤ 0.08, Comparative Fit Index (CFI) ≥ 0.90, Tucker-Lewis Index (TLI) ≥ 0.90, Goodness-of-Fit Index (GFI) ≥ 0.90.[26] The convergent validity was assessed by composite reliability (CR) and average variance extracted (AVE)). CR ≥ 0.7 indicates high internal consistency between the latent construct and its indicators.[27] AVE ≥ 0.5 shows the intended construct accounts for more than half of the variance in the indicators.[28] Discriminant validity was assessed by multiple methods to ensure distinct constructs.[29] Pearson correlations below 0.8 indicate differentiation between constructs. Fornell-Larcker criterion: the square root of the AVE for each factor must exceed its highest correlation with any other factor.[30] The Heterotrait-Monotrait (HTMT) ratio was also used, with values under 0.85 indicating clear construct distinction.[31].

To respond to the second objective, the Mann-Whitney and Kruskal Wallis tests were used to determine differences among student groups classified by gender, age, field of study, and academic year regarding the level of digital competence. For significant Kruskal-Wallis results, pairwise post-hoc comparisons were performed using the Bonferroni correction to control for type I error. Additionally, effect sizes were calculated to assess the practical significance of differences. For Mann-Whitney U tests, the

effect size was computed as r = Z/ $\sqrt{}$N, where Z is the test statistic and N is the total number of observations. For Kruskal-Wallis tests, the effect size was computes as epsilon squared (ε^2)

$$\varepsilon^2 = \frac{H - k + 1}{N - k}$$

Where H = Kruskal-Wallis test statistic; k = number of groups; N = total number of observations.

Non-parametric tests (Mann-Whitney and Kruskal-Wallis) were applied as they are appropriate for Likert-type response data, and group comparisons followed a basic consistency check to ensure valid interpretation.

Sampling and data collection

The stratified sampling method used a list of all students in the college's four majors: pharmacy, nursing, lab medical technology, and physical therapy. Students from each major were selected randomly, with the number of students chosen from each discipline proportional to the total number of students in that discipline within the college. The sample consisted of 750 health professions students in Dong Thap province, Vietnam. Data collection occurred from June to August 2022. Participation was voluntary, with informed consent obtained. Students were guaranteed anonymity and confidentiality in order to minimize response bias, and data collection took place in a non-evaluative environment free from academic consequences. Questionnaires were completed during class sessions, supporting a high response rate. Ultimately, 717 students participated, yielding a 95.6 % response rate.

The sample was randomly divided into two nearly equal subgroups: EFA (n = 366; 97.6 %) and CFA (n = 351; 93.6 %). This sample exceeds the commonly recommended minimum of 5–10 participants per item for factor analysis [27] and meets the threshold of n > 300 considered "good to excellent" for psychometric studies[31].

Ethics considerations

The study was conducted in accordance with the Declaration of Helsinki and was approved by the Research Ethics Review Committee for Research Involving Human Subjects of Dong Thap Medical College (Reference Code: CDYT No. 01/257, approved on June 1, 2022). Informed consent was obtained from all participants during face-to-face interactions, where the purpose and method of data collection were explained, and confidentiality of the data was ensured. All methods were carried out in accordance with relevant guidelines and regulations.

Results

Phase I

The literature review generated a digital competence framework including 43 items. Five dimensions emerged from the inductive process as shown in Table 1: (i) Information and Data Handling (9 items); (ii) Communication and Collaboration (11 items); (iii) Digital Content Creation (6 items); (iv) Digital Safety (11 items); and (v) Problem-Solving, Device Usage, and Software Operations (6 items).

Phase II

The panel of experts participated in evaluating the content validity of the instrument. Most experts provided additional comments on specific items and the scale. I-CVI of the items varied between 0.79 and 1. According to the evaluation and written feedback provided by the expert panel, items with I-CVI scores of less than 0.83 were revised or deleted. As a result, some minor modifications were made, such as item 16: "I consistently share high-quality, original, and valuable content that is beneficial to the wider audience.", and item 18: "My online contributions consistently consist of excellent, unique, and valuable posts or articles that benefit the public.", integrated to create an amendment of item 16: "I post only good, original, unique, and useful posts/articles that are useful to the public." In addition, five items with I-CVI scores less than 0.83 were deleted, following the recommendations of experts and the research team's decision. Ten health professions students took part in the pilot study to assess the items' suitability and explicitness. The pilot test's feedback led to some modest rewording of eight items (Information and data handling, three items; communication and collaboration, two items; and Digital content creation, three items). The dataset used for instrument validation did not contain the pilot study's responses. After completely editing the items based on expert feedback and pilot testing, the first version of the DigiCAS-HPS included 37 items in Supplement 1.1/Questionnaire 1, with 32 items achieving an I-CVI score of 1 and five items receiving an I-CVI score of 0.83. The S-CVI was 0.98 in Table A1/Supplement 2.

Phase III

In the total sample, the mean age of students was 21.7 years (SD 5.45), with 76.63 % female students. Table 2 showed most respondents were enrolled in their first year of study (48.9 %) and used smartphones as their preferred internet device (95.2 %). Before EFA, the quality and factorability of the data were assessed. No missing data to handle was required since all items were mandatory. Multivariate normality has been tested by comparing Mardia's kurtosis coefficient (112.38), which is defined by the mean of the squared Mahalanobis distances with a threshold value (number of items multiplied by the number of items plus two) (1443). Mardia's kurtosis coefficient being less than a threshold value suggested that the data did not deviate strongly from multivariate normality regarding kurtosis.[32].

The EFA was used to identify the underlying latent factor structure of the first version of the DigCAS-HPS (37 items) in Fig. 1/ Supplement 3. Eleven low and cross-loading items (factor loadings < 0.65) were removed. Table 3 revealed the second verison of DigiCAS-HPS represented three factors with 26 items. The three-factor solution was validated by the scree plot presented in Fig. 1. In a scree plot, it is preferable to see a dramatic decline in eigenvalue size, with the remaining smaller eigenvalues being debris. When the eigenvalues decrease significantly, the information recovered by an extra factor would be comparatively minimal.

The total variance explained by the second version of DigiCAS-HPS was 51.60 % (>50 % threshold).[33] KMO test (0.94) confirmed the sampling adequacy and BTOS (p < 0.001) revealed that correlations between items existed in the factor model.

In order to evaluate the goodness-of-fit and convergent and discriminant validity of the second version of DigiCAS-HPS, CFA was carried out based on the EFA factor model. The result of the assessment of the goodness-of-fit in Table 4 was shown according to the following model fit indices.

The null hypothesis was the factor model of second version of DigiCAS-HPS fitted observed data. In Table 4, $\chi 2/df$ = 2.681, GFI = 0.847, and Root Mean Square Error of Approximation (RMSEA) = 0.069 (p-value < 0.0001) showed the null hypothesis of model fit was rejected. The relative fit indices included Comparative Fit Index (CFI) = 0.89, and Tucker-Lewis Index (TLI) = 0.88 revealed the model and data did not fit well.

Table 1
The identification of domains in the digital competence framework resulted from the literature review.

Domains of scale Literature sources	Information and data handling (9 items)	Communication and collaboration (11 items)	Digital content creation (6 items)	Digital safety (11 items)	Problem –solving and device usage and software operations (6 items)
Digital competence framework European Commission[12]	Browsing, searching and filtering data, information and digital content Evaluating data, information and digital content Managing data, informationand digital content	Interacting through digital technologies Sharing through digital Technologies Engaging in citizenship through digital technologies. Collaborating through digital technologies	Developing digital content Integrating and re-elaborating digital contentDigital responsibility	Protecting personal data and privacy	Solving technical problems Creatively using digital technologiesProgramming
UK Department for Education[13]	Finding and evaluating Information Managing and storinginformation	Communicating and sharing	Creating and editing documentsCreating and editing digital media	Protecting dataBeing responsible online	Using devices Identifying and solvingtechnical problems
Fraillon et al[14]	Accessing and evaluating information Managing information	Sharing information	Transforming informationCreating information	Using informationresponsibly and safely	Foundations of computer use
Systematic reiew Konttila et al[15] Nazeha et al [9] Jimenez et al[10]	Knowledge of digital technology and the digital skills required to provide good patient care. Access and manage digital health information Knowledge and skills in computer basics, information systems Knowledge and skills for technologies to support the management and administration of primary care practice. Knowledge and skills to search and evaluate online health information. Skills related to use of technologies to analyse data. Knowledge of basic medical informatics, concepts and principlesrelated to information	Social and communication skills associated with digitalisation in patient care. Digital communications to enhance interpersonal skills and decision-making processes. Telecommunications, data exchange: knowledge and skills related to internet, email, databases, remote maintenance, connection to hospitals, andreferrals	Knowledge and use of digital health technologies for medical education and teaching	Ethical considerations of digitalisation in patient care Privacy and security for digital data and health information Knowledge of privacy and security regulations of health-related data Knowledge of legislation or regulations related to technologies or digital health dat	Technical knowledge of ICT systems, including software applications Knowledge and skills for basic use of computers, tablets, and smartphones
Qualitative studies Mikkonen[16]		Competence in interacting with the patient throughdigital means (video channels)			Competence in using the computer, digital equipment and digital programmes.Know how to access ICT support and to solve technical problems

We improved model fit based on modification indices (MI). Firstly, items (Q1, Q4, Q5, Q9, Q23, Q29, Q33, and Q34) with standardized residual covariances greater than 2 were removed. Therefore, factor F3 (digital safety) was decided to be removed because it only included 2 items (Q35 and Q37), which did not meet the general rule of having at least 3 items per factor.[34] Next, covariances were added between e10 and e5, e4 and e5 due to the large modification indices (MI > 20). After removing a total of ten items, the final version of DigiCAS-HPS resulted in two factors in Fig. 2: Factor F1 (Digital Interaction & Responsibility) and Factor F2 (Digital Content Development and Software Mastery) with 16 items. The details of 16 items in the DigiCAS-HPS were includes in Supplement 1.2 /Appendix A.

Overall, the fit indices of this version of the model were within acceptable levels as shown in Table 5. Among the absolute fit indices, the chi-square value standardized by degrees of freedom ($\chi 2/df$) was 1.94 < 3 (acceptable fit). Furthermore, GFI was 0.94, and RMSEA was 0.05 with a p-value of 0.375, indicating that the model fitted the observed data well. Combining the relative fit indices, CFI and TLI were also equal to 0.96, and SRMR was 0.04. These values might confirm a perfect fit to the data.

In order to confirm the reliability and validity of the final version of DigiCAS-HPS (16 items), the analysis results of internal consistency reliability, convergent validity and discriminant validity were as follows:

In Table 6, the overall Cronbach's alpha, and Cronbach's alpha of each factor exceeding 0.8, indicated that the final model had strong internal consistency. Observed items within each factor and across all factors were measuring the intended constructs effectively and consistently.

Table 2
Description of participant's characteristics.

Characteristics	EFA group N (%)	CFA group N (%)	Total N(%)
N	366	351	717
Gender			
Male	88 (24.0)	82 (23.36)	170 (24)
Female	278 (76.0)	269 (76.63)	547 (76)
Year of study			
1st year	191 (52.2)	160 (45.58)	351 (48.9)
2nd year	77 (21.0)	81 (23.07)	158 (21)
3rd year	98 (26.8)	110 (31.3)	208 (26.8)
Major of study			
Pharmacy	206 (56.3)	204 (58.1)	404 (56.3)
Nursing	131 (35.8)	120 (34.2)	257 (35.8)
Medical Lab Technology	15 (4.1)	15 (4.3)	29 (4.1)
Physiotherapy	14 (3.8)	12 (3.4)	27 (3.8)
Internet device use			
Desktop	8 (2.2)	8 (2.3)	16 (2.2)
Laptop	9 (2.5)	9 (2.5)	18 (2.5)
Smartphone	349 (95.35)	334 (95.15)	683 (95.2)
Age			
Mean ± SD	21.9 ± 5.7	21.5 ± 5.12	21.7 ± 5.45

Convergent validity was tested by examining the standardized factor loadings of items and values of AVE and CR. All standardized factor loadings of items exceeding 0.6 indicated a high convergent validity. AVE of Factor 1 (F1) and Factor 2 (F2) met the criterion (AVE $\geq$ 0.5). Both factors had acceptable convergent validity because each factor might explain at least 50 % of item variance. The CR statistics of factors ranged from 0.84 to 0.92 and were above the threshold of 0.7. Both AVE values and CR statistics provided evidence of the convergent validity of the final version of DigiCAS-HP.

Discriminant validity between the two factors was assessed based on the following criteria: (i) The correlation coefficient between factors F1-F2 (0.63) was smaller than the square root of AVE for each factor F1 (0.71) and F2 (0.72) in Table A2/Supplement 2. It demonstrated that the constructs F1 and F2 were distinct and not closely related; (ii) Furthermore, the heterotrait-monotrait ratio (HT-MT) equal to 0.63 < 0.85 in Table A3/ Supplement 2 revealed the two constructs were indeed distinguishable from each other.[35] The results of goodness-of-fit, reliability and construct validity emphasized the robustness of the final version of Digi-CAS-HPS scale and reinforced the overall validity of the measurement model.

Differences of digital competence among health professions students assessed by DigiCAS-HPS scale

Table 7 indicated that older students and students nearing graduation tended to demonstrate higher levels of proficiency in digital content development and software mastery, and possessed more advanced competencies in these areas, respectively. Effect sizes ranged from small to medium, indicating practical differences in digital skill mastery across these groups (Appendix A, Supplement 2: Table A.4).

Discussion

According to the World Health Organization, integrating digital technology into health has transformed healthcare by improving the quality, affordability, and accessibility of health services.[36,37]. In the United Kingdom, 90 % of the National Health Service positions expect to require digital capabilities within the next 20 years.[38] This shift underscores the critical role of eHealth—the use of information and communication technologies in healthcare— and digital capabilities in shaping the future of healthcare delivery. [39] It reveals the growing demand for a digitally competent healthcare workforce in the future who can effectively adopt and integrate eHealth technologies into clinical and educational settings.[38].

To clarify the necessity of the DigiCAS-HPS in assessing essential competencies on integrating eHealth technologies into routine healthcare practices, we discuss the significance of measurement scales in terms of eHealth aspects and the demands of technology-driven healthcare systems.

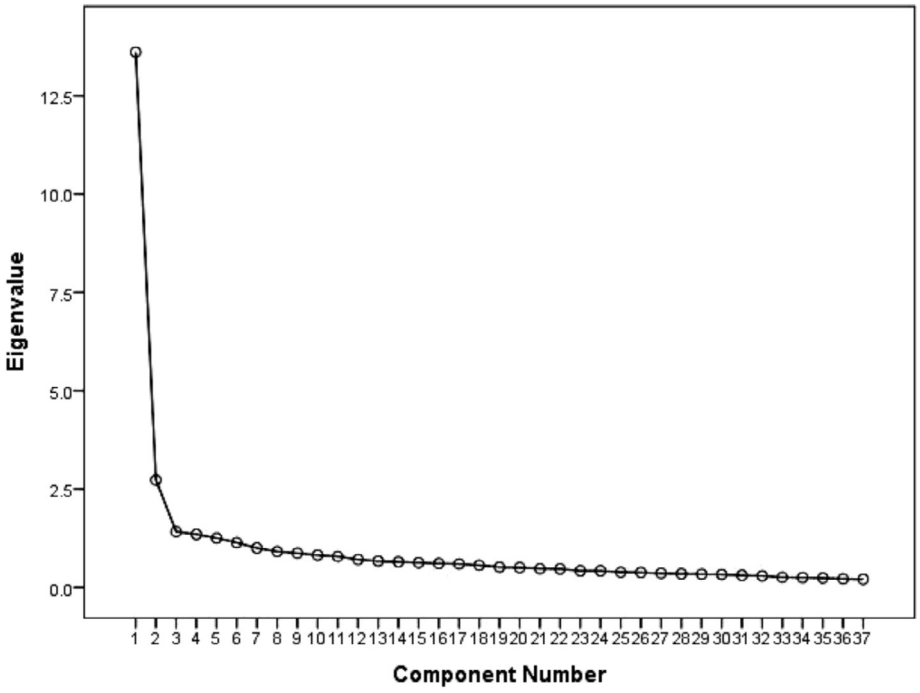


Fig. 1. Scree plot of the first version of DigiCAS-HPS (37 items).

Table 3
EFA-based factor loadings and internal consistency reliability in the second version of DigiCAS-HPS (26 items).

Second version of DigiCAS-HPS items	Factor loadings[a]		
Factor 1: Digital information literacy and responsibility	1	2	3
13 items			
Q1	0.66		
Q3	0.74		
Q5	0.68		
Q6	0.68		
Q8	0.7		
Q15	0.78		
Q16	0.72		
Q17	0.71		
Q19	0.7		
Q20	0.67		
Q21	0.81		
Q22	0.75		
Q31	0.73		
Factor 2: Digital content creation and software mastery			
8 items			
Q2		0.74	
Q4		0.68	
Q9		0.81	
Q10		0.77	
Q13		0.8	
Q28		0.71	
Q29		0.72	
Q36		0.72	
F3: Digital safety			
5 items			
Q23			0.67
Q33			0.71
Q34			0.68
Q35			0.67
Q37			0.73
Eigenvalue	13.61	2.73	1.42
Percentage of variance	39.65	7.93	4.02
Total percentage of factor model	51.6		
Cronbach's alpha	0.92	0.9	0.79
Cronbach's alpha on total scale	0.94		

[a] Extraction method: Principal Component Analysis. Rotation Method: ProMax with Kaiser Normalization

Table 4
Model fit indices of the second version of DigiCAS-HPS – EFA factor model (26 items).

Model fit indices	Second version of DigiCAS-HPS	Model fit thresholds
Absolute fit indices		
Chi – Square (χ^2)	793.612	
Chi – Square/DF (χ^2/df)	2.681	≤ 3: acceptable fit
RMSEA (Root-mean-square error of approximation)	0.069	≤ 0.05: excellent fit
RMSEA p- value	p < 0.0001	>0.05: acceptable fit
GFI (Goodness-of-Fit index)	0.847	≥ 0.95: excellent fit ≥ 0.9: acceptable fit
Relative fit indices		
SRMR (Standardized root-mean-square residual)	0.064	≤ 0.08: acceptable fit
TLI (Tucker – Lewis index)	0.88	>0.95: good fit
CFI (Comparative fit index)	0.89	≥ 0.95: excellent fit ≥ 0.9: acceptable fit

The DigiCAS-HPS resulted in 16 items with a refined two-factor structure. Factor 1: Digital Interaction & Responsibility reflects a comprehensive set of competencies essential for functional and

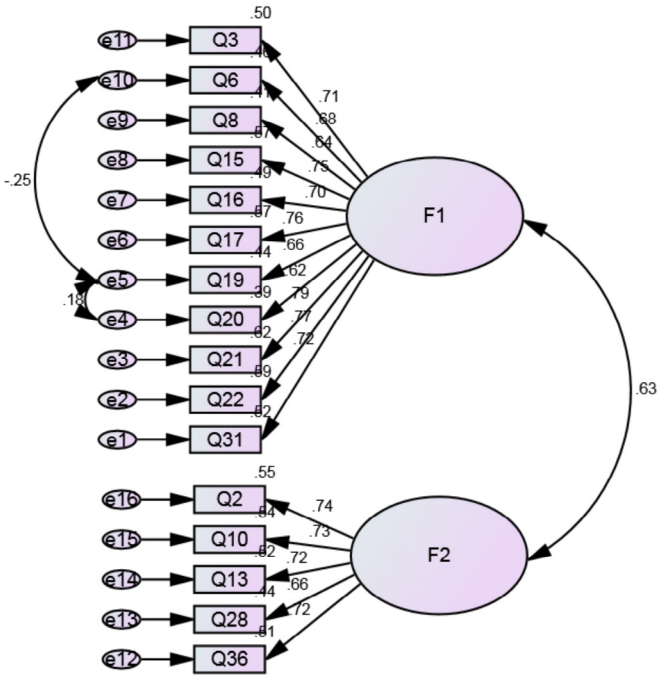


Fig. 2. The final version of DigiCAS-HPS.

Table 5
Model fit indices of the final version of DigiCAS-HPS – CFA factor mode (16 items).

Model fit indices	Final version of DigiCAS-HPS	Model fit thresholds
Absolute fit indices		
Chi – Square (χ^2)	196.07	
Chi – Square/DF (χ^2/df)	1.94	≤ 3: acceptable fit
RMSEA (Root-mean-square error of approximation)	0.05	≤ 0.05: excellent fit
RMSEA p- value	p = 0.375	>0.05: acceptable fit
GFI (Goodness-of-Fit index)	0.94	≥ 0.95: excellent fit ≥ 0.9: acceptable fit
Relative fit indices		
SRMR (Standardized root-mean-square residual)	0.04	≤ 0.08: acceptable fit
TLI (Tucker – Lewis index)	0.96	>0.95: good fit
CFI (Comparative fit index)	0.96	≥ 0.95: excellent fit ≥ 0.9: acceptable fit

ethical engagement with digital tools in health education and practice. This factor integrates four thematic domains—digital communication, collaboration, health literacy, and online responsibility and security—which together support students' ability to engage in technology-mediated teamwork, knowledge exchange, and safe information practices. These skills are increasingly relevant for virtual care coordination, evidence-based decision-making, and participation in online communities of practice, all of which are foundational in eHealth environments. [40–45] Ramsden et al. highlighted the value of online platforms for fostering professional confidence, enabling knowledge sharing, and overcoming geographic barriers—competencies that are foundational not only for current students but also for future digital healthcare workers in underserved areas [40]. Siddiq et al. found that students who demonstrated

Table 6
Convergent validity and internal consistency reliability of final version of DigiCAS-HPS.

Factor	Standardizedfactor loadings	AVE (≥ 0.5)	CR (>0.7)	Cronbach's alpha (>0.7)
Digital interaction and responsibility (F1)		0.50	0.92	0.914
Q3	0.71			
Q6	0.68			
Q8	0.64			
Q15	0.76			
Q16	0.70			
Q17	0.76			
Q19	0.66			
Q20	0.62			
Q21	0.79			
Q22	0.77			
Q31	0.72			
Digital content development and software mastery (F2)		0.51	0.84	0.84
Q2	0.74			
Q10	0.73			
Q13	0.72			
Q28	0.66			
Q36	0.72			
Final version of DigiCAS-HPS (16 items)				0.912

Table 7
Differences in Digital Interaction & Responsibility (F1) and Digital Content Development and Software Mastery (F2) between students groups of demographic and academic characteristics.

Grouping variables	F1	F2
Age: 18, 19–20, >20 years od		
Chi-square	2.34	35.002
p-value [a]	0.31	< 0.001
Academic years: first, second, third & fourth year		
Chi-square		
p-value [a]	0.683	15.197
	0.711	0.001
Field of study: Pharmacy, nursing, lab technology, physiotherapy		
Chi-square		
p-value [a]		
	2.578	1.807
	0.461	0.613
Gender: male, female		
Z	−0.17	−1.062
p-value [b]	0.865	0.288

[a] p-value from Kruskal Wallis test; [b] p-value from Mann-Whitney test.

strong abilities in information management, ICT-based communication and collaboration, and critical use of digital tools were more likely to be digitally prepared for academic and professional environments. [46] Thus, DigiCAS-HPS may serve as an essential tool for evaluating preparedness for digitally enabled professional learning and collaboration in both academic and real-world health contexts.

The another scoping review on Virtual Communities of Practice (VCoPs) in healthcare emphasized digital communication, collaboration, trust, and technological competence as critical to successful professional networks. [43,44] The DigiCAS-HPS items assessing students' use of online tools, ethical behavior, and content sharing mirror the competencies required for effective participation in VCoPs—settings. Notably, factor F1 aligns with the DigComp 2.2 framework, which emphasizes not only technical proficiency but also ethical, critical, and collaborative digital behavior. [12] Given the growing integration of digital technologies in healthcare, ethical awareness of data privacy and cybersecurity has become a critical dimension of digital literacy. The competencies in F1, such as using licensed software, updating passwords, and considering the appropriateness of sharing personal data online, are directly assessed. Our findings reinforce that promoting cybersecurity awareness through education is essential to preparing future health professionals to safeguard sensitive patient data and adhere to legal and ethical standards. This aligns with recent reviews identifying user behavior and digital hygiene as core to cybersecurity resilience in healthcare.[47] Therefore, DigiCAS-HPS can serve as both an assessment and an educational tool for fostering ethically grounded digital practice.

Factor 2: Digital Content Development and Software Mastery reflects the necessary skills for creating, organizing, and managing digital information, and fundamental competencies for operating and maintaining digital tools and systems in eHealth environments. The skills listed in Factor 2 are essential for working with new digital health systems like Electronic Health Records (EHRs) and health information systems that can share data with each other. As outlined by Pan American Health Organization, the COVID-19 pandemic pointed out the urgent need for healthcare professionals to efficiently create, organize, and manage digital information to support timely decision-making, continuity of care, and public health surveillance.[48] The ability to operate digital tools and ensure proper formatting, data integrity, and platform navigation are no longer technical distinctions but essential skills for real-time data exchange, collaborative diagnosis, and clinical decision support. EHR interoperability, which enables patient data to move easily between different systems and institutions, depends on users being skilled in organizing digital information and using software—skills that are specifically covered in Factor 2 of the DigiCAS-HPS. Next, the software mastery domain aligns with core recommendations from cybersecurity literature. A systematic review of modern threats and trends in healthcare cybersecurity identified unsafe software use, outdated systems, and poor user training as major contributors to healthcare cyber breaches.[47] Skills assessed in F2, such as responsible use of licensed software and regular password updates, reflect critical cybersecurity behaviors.

In summary, factor F2 reflects educational digital proficiency and aligns with the operational requirements of digitally integrated healthcare delivery post-pandemic. It not only measures technical proficiency but also serves as a proxy for evaluating the foundational digital security practices required in modern eHealth systems.

These results highlight the tool's utility in assessing key dimensions of digital competence relevant to educational and clinical contexts. While DigiCAS-HPS does not directly assess competencies in emerging technologies such as artificial intelligence, blockchain, or clinical decision support systems, it captures foundational digital skills—such as responsible online behavior, content cre-

ation, and collaboration—that are essential prerequisites for effectively engaging with and adapting to these advanced digital health tools.

To address the second objective, we applied the DigiCAS-HPS to assess differences in digital competence across groups of students. The analytic results show the older students were more proficient with software and digital content creation. It might explain the possible influence of age and cumulative experience on students' digital growth. We also identified that students nearing graduation displayed more advanced competencies in these areas. Advanced coursework and clinical practice during their studies might increase the possibility of a trend toward greater exposure to digital technologies. The significant differences in digital skills across students at different academic levels illustrate this trend and emphasize the impact of curriculum integration and educational advancement on the development of digital skills. However, no statistically significant differences were observed in DigiCAS-HPS scores across gender, suggesting that male and female health professions students demonstrate comparable levels of digital competence. This finding aligns with previous research showing decreasing gender disparities in ICT-related competencies and attitudes. For example, the TEDDICS study found no gender differences in the emphasis on digital skills, supporting the idea that modern digital literacy constructs function equivalently across gender groups.[46] Similarly, other studies also reported no significant gender effects on ICT-related attitudes or behaviors.[49–51] These converging findings suggest that increasing access to digital technologies and evolving educational practices may have contributed to the narrowing of the gender gap in digital competencies. While earlier studies noted gender-based discrepancies [52,53], more recent evidence, including the TALIS 2013 report, highlights the trend toward greater parity.[54] Thus, the lack of significant gender differences in DigiCAS-HPS scores likely reflects this broader shift, and supports the scale's validity and generalizability across diverse student cohorts.

Limitation

The focus on health professions students within a single institutional type aligns with the study's objective of developing a scale tailored specifically to assess the digital competencies of future health professionals. This deliberate scope ensures the scale addresses the unique educational needs of health professions students, as they represent the future healthcare workforce. However, the scale has not been validated in other institutions or contexts, such as those outside the health sector or among practicing health staff, which may limit its applicability to those broader populations. Nevertheless, this targeted approach enhances its relevance and utility for tailoring education within the health sector. Moreover, the DigiCAS-HPS relies on self-reported responses, which may introduce social desirability or recall biases—potentially leading to over- or underestimation of actual competencies. To enhance measurement validity, future studies could incorporate complementary performance-based assessments or observational methods that evaluate real-time use of digital tools in simulated or clinical environments. This multimethod approach could offer a more comprehensive understanding of digital competence in action. Additionally, as digital literacy is influenced by educational, cultural, and infrastructural contexts, follow-up research is warranted to explore the scale's applicability and measurement invariance across different countries, regions, or professional domains, including in-service healthcare workers and interprofessional teams. Cross-cultural validation would strengthen the generalizability of the DigiCAS-HPS and support its integration into global health education frameworks.

Conclusion & recommendations

The DigiCAS-HPS demonstrated robust construct validity and internal consistency across its two-factor structure: Digital Interaction & Responsibility and Digital Content Development and Software Mastery. The scale provides a reliable and evidence-based framework for assessing digital competence among health professions students, ensuring they are well-prepared for today's increasingly technology-integrated healthcare environments.

Beyond assessment, DigiCAS-HPS offers actionable value for educational curriculum design and workforce development. Healthcare education programs can integrate DigiCAS-HPS at both entry and exit points in training to benchmark and monitor students' digital skill progression. Its results can inform targeted instructional interventions, such as modules on cybersecurity, ethical digital practice, or collaborative software tools in clinical settings. For in-service professionals, healthcare institutions can employ DigiCAS-HPS in continuing professional development (CPD) programs to tailor digital upskilling efforts, ensuring alignment with evolving eHealth system requirements.

As a self-assessment and developmental tool, DigiCAS-HPS empowers learners to recognize and close personal gaps in digital competence. Future implementation across diverse contexts may support international comparisons, longitudinal studies, and refinement of digital health curricula. Ongoing research and adaptation will further strengthen its role in advancing digitally competent, ethically responsible, and workforce-ready healthcare professionals.

CRediT authorship contribution statement

Cua Ngoc Le: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Uyen Thi To Nguyen:** Visualization, Validation, Supervision, Resources, Investigation, Data curation, Conceptualization. **Luan Minh Le:** Validation, Supervision, Resources, Investigation, Data curation. **Dien Phu Tran:** Software, Resources, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

1. Clarke-Darrington J, McDonald T, Ali P. Digital capability: an essential nursing skill for proficiency in a post-COVID-19 world. *International Nursing Review*. 2023;70(3):291–296. https://doi.org/10.1111/inr.12839.
2. Li C, Lalani F. The COVID-19 pandemic has changed education forever. In World Economic Forum (Vol. 29). 2020.
3. World Health Organization. Novel coronavirus (2019-nCoV): situation report, https://apps.who.int/iris/bitstream/handle/10665/330776/nCoVsitrep31Jan2020-eng.pdf; 2020. [accessed 23 December 2023].
4. Radovanović D, ed. *Digital Literacy and Inclusion: Stories, Platforms, Communities*. Springer International Publishing; 2023:195–214. https://doi.org/10.1007/978-3-031-30808-6.
5. Nowell L, Dhingra S, Carless-kane S, et al. A systematic review of online education initiatives to develop students remote caring skills and practices. *Medical Education Online*. 2022;27(1). https://doi.org/10.1080/10872981.2022.2088049.
6. Clarke-Darrington J, McDonald T, Ali P. Digital capability: an essential nursing skill for proficiency in a post-COVID-19 world. *International Nursing Review*. 2023;70(3):291–296. https://doi.org/10.1111/inr.12839.
7. Jarva E, Oikarinen A, Andersson J, et al. Healthcare professionals' perceptions of digital health competence: a qualitative descriptive study. *Nursing Open*. 2022;9(2):1379–1393. https://doi.org/10.1002/nop2.1184.
8. Suonnansalo P, Kaihlaniemi J, Kähkönen O, Oikarinen A. Patients' perspectives of the digital counselling competence of healthcare professionals—A qualitative descriptive study. *Journal of Clinical Nursing*. 2024;33(11). https://doi.org/10.1111/jocn.17354.
9. Nazeha N, Pavagadhi D, Kyaw BM, Car J, Jimenez G, Car LT. A digitally competent health workforce: scoping review of educational frameworks. *JMIR.*. 2020;22(11). https://doi.org/10.2196/22706.
10. Jimenez G, Spinazze P, Matchar D, et al. Digital health competencies for primary healthcare professionals: a scoping review. *International Journal of Medical Informatics*. 2020;143. https://doi.org/10.1016/j.ijmedinf.2020.104260 104260.
11. Jarva E, Oikarinen A, Andersson J, Tomietto M, Kääriäinen M, Mikkonen K. Healthcare professionals' digital health competence and its core factors; development and psychometric testing of two instruments. *International Journal of Medical Informatics*. 2023;171. https://doi.org/10.1016/j.ijmedinf.2023.104995 104995.
12. European Commission. Digcomp Framework 2.2, https://joint-research-centre.ec.europa.eu/digcomp/digcomp-framework_en; 2023 [accessed February 22, 2023].
13. UK Department for Education. National standards for essential digital skills; 2019 [accessed May 10, 2023] https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/796596/National_standards_for_essential_digital_skills.pdf.
14. Fraillon J, Ainley J, Schulz W, Duckworth D, Friedman T. IEA international computer and information literacy study 2018 assessment framework. *Springer Nature*. 2019. https://doi.org/10.1007/978-3-030-19389-8.
15. Konttila J, Siira H, Kyngäs H, et al. Healthcare professionals' competence in digitalisation: a systematic review. *Journal of Clinical Nursing*. 2018;28(5–6):745–761. https://doi.org/10.1111/jocn.14710.
16. Jarva E, Oikarinen A, Andersson J, et al. Healthcare professionnals' perceptions of digital health competence: a qualitative descriptive study. *Nursing Open*. 2022;9(2):1379–1393. https://doi.org/10.1002/nop2.1184.
17. Jarva E, Mikkonen K, Andersson J, et al. Aspects associated with health care professionals' digital health competence development – a qualitative study. *FinJeHeW.*. 2022;14(1):79–91. https://doi.org/10.23996/fjhw.111771.
18. Mikkonen K, Tomietto M, Watson R. Instrument development and psychometric testing in nursing education research. *Nurse Education Today*. 2022;119. https://doi.org/10.1016/j.nedt.2022.105603 105603.
19. Polit DF, Beck CT. Focus on research methods. is the CVI an acceptable indicator of content validity? Appraisal and recommendations. *Research in Nursing & Health*. 2007;30(4):459–467. https://doi.org/10.1002/nur.20199.
20. Polit DF, Beck CT. The content validity index: are you sure you know what's being reported? Critique and Recommendations. *Research in Nursing & Health*. 2006;29(5):489–497. https://doi.org/10.1002/nur.20147.
21. Bannigan K, Watson R. Reliability and validity in a nutshell. *Journal of Clinical Nursing*. 2009;18(23):3237–3243.
22. Briggs NE, MacCallum RC. Recovery of weak common factors by maximum likelihood and ordinary least squares estimation. *Multivariate Behavioral Research*. 2010;45(1):171. https://doi.org/10.1080/00273170903504810.
23. Cudeck R. Exploratory factor analysis. In: Tinsley HEA, Brown SD, eds. *Handbook of Applied Multivariate Statistics and Mathematical Modeling*. Elsevier; 2000:265–296.
24. Stevens J. *Applied multivariate statistics for the social sciences*. Mahwah, NJ: Lawrence Erlbaum Associates; 2002.
25. DeVellis RF, Thorpe CT. *Scale development: Theory and applications*. 5th ed. Sage Publications; 2021.
26. Hu LT, Bentler PM. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Structural Equation Modeling*. 1999;6(1):1–55.
27. Hair JF, Babin BJ, Black B, Anderson RE, Tatham RL. *Multivariate data analysis: a global perspective*. 7th ed. Pearson Education; 2010.
28. Streiner D. 10 figuring out Factors: the use and Misuse of factor Analysis*A Guide for the Statistically Perplexed: Selected Readings for Clinical Researchers*. Toronto: University of Toronto Press; 2012:110–122. https://doi.org/10.3138/9781442662827-012.
29. Hair JF, Black WC, Babin BJ, Anderson RE. *Multivariate data analysis*. 6th ed. Pearson Prentice Hall; 2006.
30. Henseler J, Ringle CM, Sarstedt M. A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*. 2015;43(1):115–135. https://doi.org/10.1007/s11747-014-0403-8.
31. MacCallum RC, Roznowski M, Mar C, Reith JV. Alternative strategies for cross-validation of covariance structure models. *Multivariate Behavioral Research*. 1994;29(1):1–32. https://doi.org/10.1207/s15327906mbr2901_1.
32. Tabachnick BG, Fidell LS. Cleaning up your act: Screening data prior to analysis. In: Using multivariate statistics. 5th ed. Boston: Allyn & Bacon; 2001. p. 61-116.
33. Streiner DL. Figuring out factors: the use and misuse of factor analysis. *Canadian Journal of Psychiatry*. 1994;39(3):135–140. https://doi.org/10.1177/070674379403900303.
34. Hair JF, Black WC, Babin BJ, Anderson RE. *Multivariate data analysis*. 7th ed. Pearson Prentice Hall; 2010.
35. Henseler J, Ringle CM, Sarstedt M. A new criterion for assessing discriminant validity in variance-based structural equation modeling. *JAMS*. 2015;43:115–135. https://doi.org/10.1007/s11747-014-0403-8.
36. World Health Organization Guideline. Recommendations on digital interventions for health system strengthening. World Health Organization. 2019:2020-10.
37. Areru HA, Dangisso MH, Lindtjørn B. Low and unequal use of outpatient health services in public primary health care facilities in southern Ethiopia: a facility-based cross-sectional study. *BMC Health Services Research*. 2021;21:1–4. https://doi.org/10.1186/s12913-021-06846-x.
38. Topol E. The topol review. Preparing the healthcare workforce to deliver the digital future. 2019:1-48.
39. Health Service Executive. Digital Health Capability Building Overview [Internet]. eHealth Ireland. Available from: https://www.ehealthireland.ie/technology-and-transformation-functions/digital-for-care-2030/digital-health-capability-building/digital-health-capability-building-overview/.
40. Ramsden R, Colbran R, Christopher E, Edwards M. The role of digital technology in providing education, training, continuing professional development and support to the rural health workforce. *Health Education*. 2022;122(2):126–149.
41. McTighe AJ, Donovan S. Group dynamics in healthcare settings: a strategic framework for cohesion. *RPBS.*. 2017;5(1):30–35. https://doi.org/10.12691/rpbs-5-1-5.
42. Noar AP, Jeffery HE, Subbiah Ponniah H, Jaffer U. The aims and effectiveness of communities of practice in healthcare: a systematic review. *PLoS One*. 2023;18(10). https://doi.org/10.1371/journal.pone.0292343 e0292343.
43. Shaw L, Jazayeri D, Kiegaldie D, Morris ME. Implementation of virtual communities of practice in healthcare to improve capability and capacity: a 10-year scoping review. *IJERPH.*. 2022;19(13):7994. https://doi.org/10.3390/ijerph19137994.
44. Ranmuthugala G, Plumb JJ, Cunningham FC, Georgiou A, Westbrook JI, Braithwaite J. How and why are communities of practice established in the healthcare sector? a systematic review of the literature. *BMC Health Services Research*. 2011;11:273.
45. Arias López MD, Ong BA, Borrat Frigola X, et al. Digital literacy as a new determinant of health: a scoping review. *PLOS Digital Health*. 2023;2(10). https://doi.org/10.1371/journal.pdig.0000279 e0000279.
46. Siddiq F, Scherer R, Tondeur J. Teachers' emphasis on developing students' digital information and communication skills (TEDDICS): a new construct in 21st century education. *Computers & Education.*. 2016;1(92):1–4. https://doi.org/10.1016/j.compedu.2015.10.006.
47. Kruse CS, Frederick B, Jacobson T, Monticone DK. Cybersecurity in healthcare: a systematic review of modern threats and trends. *Technology and Health Care*. 2017;25(1):1–10. https://doi.org/10.3233/THC-161263.
48. Pan American Health Organization. Digital health: Electronic health records and interoperability [Internet]. [cited 2024 Dec 15]. Available from: https://iris.paho.org/bitstream/handle/10665.2/52003/Factsheets-Digital_Health-EHR-Interoperability-eng.pdf.
49. Antonietti A, Giorgetti M. Teachers' beliefs about learning from multimedia. *Computers in Human Behavior*. 2006;22(2):267–282. https://doi.org/10.1016/j.chb.2004.06.002.
50. Sang G, Valcke M, Van Braak J, Tondeur J. Student teachers' thinking processes and ICT integration: Predictors of prospective teaching behaviors with educational technology. *Computers & Education*. 2010;54(1):103–112. https://doi.org/10.1016/j.compedu.2009.07.010.
51. Shapka JD, Ferrari M. Computer-related attitudes and actions of teacher candidates. *Computers in Human Behavior*. 2003;19(3):319–334. https://doi.org/10.1016/S0747-5632(02)00059-6.
52. Shashaani L. Gender-based differences in attitudes toward computers. *Computers & Education*. 1993;20(2):169–181. https://doi.org/10.1016/0360-1315(93)90085-W.
53. Sink C, Sink M, Stob J, Taniguchi K. Further evidence of gender differences in high school–Level computer literacy. *Chance*. 2008;21(1):49–53. https://doi.org/10.1007/s00144-008-0011-9.
54. Organisation for Economic Co-operation and Development. TALIS 2013 results: An international perspective on teaching and learning. OECD; 2014.